

What is the difference between:

- CMD & ENTRYPOINT
- COPY & ADD

- CMD and Entrypoint both tells what container has to execute when building image
- Key difference is what user can overwrite in runtime
 - CMD: can be easily overridden by user when running container
 - EntryPoint: hard to override
- Copy: simply transference files from my host machine into the container AKA "COPY"
- ADD: Does the same as copy but has more features such as :
 - Download remote urls
 - Auto extract local archives

Problem 1:

- Run the container hello-world
- Check the container status
- Start the stopped container
- Remove the container
- Remove the image

```
C:\Users\basil>docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:452a468a4bf985040037cb6d5392410206e47db9bf5b7278d281f94d1c2d0931
Status: Downloaded newer image for hello-world:latest
```

Hello from Docker!

This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:

1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
(amd64)
3. The Docker daemon created a new container from that image which runs the executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it to your terminal.

To try something more ambitious, you can run an Ubuntu container with:

```
$ docker run -it ubuntu bash
```

Share images, automate workflows, and more with a free Docker ID:

<https://hub.docker.com/>

For more examples and ideas, visit:

<https://docs.docker.com/get-started/>

```
C:\Users\basil>docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS              PORTS
37315c900cfe   hello-world    "/hello"                About a minute ago    Exited (0) About a minute ago
ee626500b6f7   jovial_solomon 7150b3a39203           34 hours ago        Exited (0) 34 hours ago
e7d0ee802983   quirky_banzai  postgres:15-alpine     2 weeks ago         Exited (137) 35 hours ago
662ba02f86f7   tuwc_postgres  docker.n8n.io/n8nio/n8n 4 months ago        Exited (0) 2 months ago
```

```
C:\Users\basil>docker start -i 37315c900cfe
```

Hello from Docker!

This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:

1. The Docker client contacted the Docker daemon.
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```
C:\Users\basil>docker rm 37315c900cfe
37315c900cfe
```

```
C:\Users\basil>docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	N
AMES						
ee626500b6f7	7150b3a39203	"/docker-entrypoint...."	34 hours ago	Exited (0) 34 hours ago		q
uirky_banzai						
e7d0ee802983	postgres:15-alpine	"docker-entrypoint.s..."	2 weeks ago	Exited (137) 35 hours ago		t
uwc_postgres						
662ba02f86f7	docker.n8n.io/n8nio/n8n	"tini -- /docker-ent..."	4 months ago	Exited (0) 2 months ago		n
8n						

```
C:\Users\basil>docker image rm hello-world
```

Untagged: hello-world:latest

Deleted: sha256:452a468a4bf985040037cb6d5392410206e47db9bf5b7278d281f94d1c2d0931

Problem 2:

- Run container centos or ubuntu in an interactive mode
- Run the following command in the container "echo docker "
- Open a bash shell in the container and touch a file named hello-docker
- Stop the container and remove it. Write your comment about the file hello-docker
- Remove all stopped containers

```
C:\Users\basil>docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
5613ea415950   ubuntu                              "/bin/bash"             2 minutes ago Exited (0) 42 seconds ago
interesting_ardinghelli
ee626500b6f7   7150b3a39203                       "/docker-entrypoint..." 35 hours ago   Exited (0) 35 hours ago
quirky_banzai
e7d0ee802983   postgres:15-alpine                 "docker-entrypoint.s..." 2 weeks ago    Exited (137) 35 hours ago
tuwc_postgres
662ba02f86f7   docker.n8n.io/n8nio/n8n            "tini -- /docker-ent..." 4 months ago   Exited (0) 2 months ago
n8n

C:\Users\basil>docker rm 5613ea415950
5613ea415950
```

```
C:\Users\basil>docker run -it ubuntu
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
817807f3c64e: Pull complete
e4a1e8de092c: Download complete
Digest: sha256:186072bba1b2f436cbb91ef2567abca677337cfc786c86e107d25b7072f
Status: Downloaded newer image for ubuntu:latest
root@5613ea415950:/# echo "docker"
docker
root@5613ea415950:/# "echo docker"
bash: echo docker: command not found
root@5613ea415950:/# touch hello-docker
root@5613ea415950:/# exit
exit
```

- When I removed the container the file was permanently deleted to make them not get deleted I have to use docker volume

```
C:\Users\basil>docker container prune
WARNING! This will remove all stopped containers.
Are you sure you want to continue? [y/N] y
Deleted Containers:
ee626500b6f7eff42ef2bfbcb5d630559dbbf78f9cbc920069cb548a104b2af08
e7d0ee802983b47728f161622ec8677fec8183646dbd2c912edef6b4351b596d
662ba02f86f773e250494dffc9d3402da981c3b8463ce54ec9e4184ca7335a2d

Total reclaimed space: 91.7MB
```

Problem 3:

- Deploy a MySQL database called app-database. Use the mysql latest image, and use the -e flag to set MYSQL_ROOT_PASSWORD to P4sSw0rd0!. The container should run in the background.

```
C:\Users\basil>docker run -d --name app-database -e MYSQL_ROOT_PASSWORD="P4sSw0rd0!" mysql:latest
Unable to find image 'mysql:latest' locally
latest: Pulling from library/mysql
660dee12630a: Pull complete
d4c7048d1cf1: Pull complete
3a83832a6208: Pull complete
c75b6f683cbd: Pull complete
f7eb4be426a1: Pull complete
ad38d99426c5: Pull complete
7ef200aee2ac: Pull complete
fc3bc5fe2030: Pull complete
dc0ae5ec7f3c: Pull complete
4e3633ebbb73: Pull complete
32f88ec2712b: Download complete
697c144a0375: Download complete
Digest: sha256:24e450bbd24f621c71b10404c946cc9ea1cbb0e6e7464b2be2de5193dcf1d05b
Status: Downloaded newer image for mysql:latest
9246a964fca466f15b013be8bb181ef3beb1145f2d3742907259df0a68305713
```

Problem 4:

- Run the image Nginx
- Add html static files to the container and make sure they are accessible
- Commit the container with image name IMAGE_NAME

```
C:\Users\basil>docker run -d -p 8080:80 --name nx nginx
e14b3e3306187fc94039bc48df20d3a0327d8b6d5057f1a209c54cc71c1806f5
```

```
C:\Users\basil>notepad index.html
```

```
C:\Users\basil>docker cp index.html nx:/usr/share/nginx/html/index.html
Successfully copied 2.05kB to nx:/usr/share/nginx/html/index.html
```

> C localhost:8080

Welcome to my custom Nginx container!

I successfully created this in Windows and copied it over.

```
C:\Users\basil>docker commit nx IMAGE_NAME
invalid reference format: repository name (library/IMAGE_NAME) must be lowercase

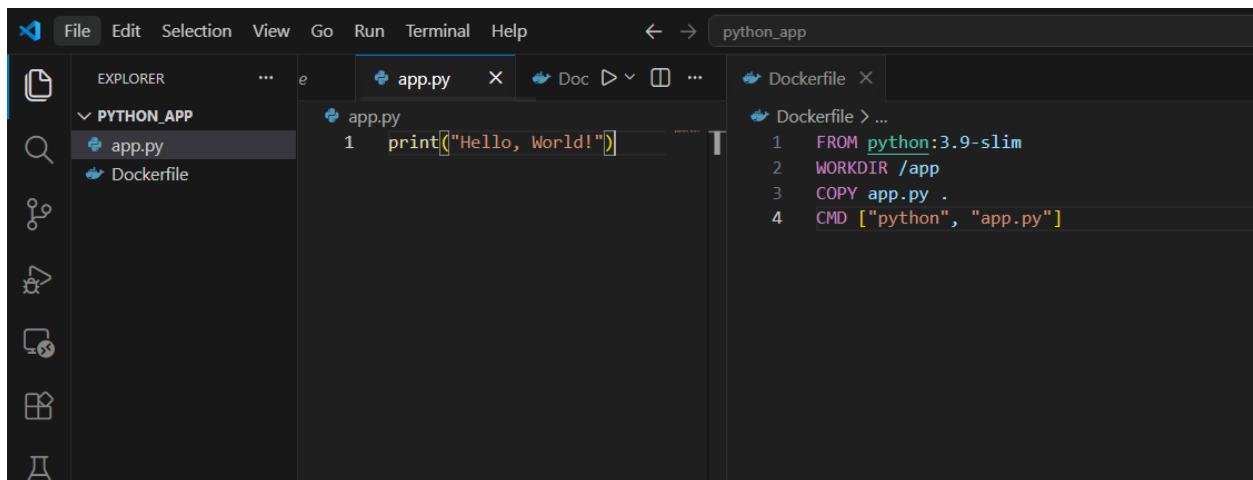
C:\Users\basil>docker commit nx image_name
sha256:0a43d41dd7fb7f1ae288e0c4419f877edfaaf50ac35f1a8875d2aaa38bc127a9

C:\Users\basil>docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
docker.n8n.io/n8nio/n8n:latest	0a65e6e5995c	1.68GB	278MB	
image_name:latest	0a43d41dd7fb	237MB	62.9MB	
mysql:latest	24e450bbd24f	1.27GB	283MB	U
nginx:latest	7150b3a39203	240MB	65.8MB	U
postgres:15-alpine	fceb6f86328c	392MB	109MB	
postgres:15.17-trixie	f30e3de0ac9c	627MB	158MB	
ubuntu:latest	186072bba1b2	119MB	31.7MB	

Problem 5:

- Create a python simple app
- Create a dockerfile to containerize the python app
- Build the image and test it
- (Bonus)create a dockerfile for the same app in smaller size using multi staging
- Push the created image into your docker hub repo



```
C:\Users\basil\Downloads\python_app>docker build -t python_app
ERROR: docker: 'docker buildx build' requires 1 argument

[+] Building 40.3s (8/9)                                docker:desktop-linux
=> [internal] load build definition from Dockerfile      0.8s
=> => transferring dockerfile: 112B                     0.0s
[+] Building 42.6s (9/9)                                docker:desktop-linux
=> [internal] load build definition from Dockerfile      0.8s
=> => transferring dockerfile: 112B                     0.0s
=> [internal] load metadata for docker.io/library/python:3.9-slim 5.3s
=> [auth] library/python:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore                        0.7s
=> => transferring context: 2B                          0.0s
=> [1/3] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13c 19.3s
=> => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf 0.8s
[+] Building 43.0s (9/9) FINISHED                        docker:desktop-linux
=> [internal] load build definition from Dockerfile      0.8s
=> => transferring dockerfile: 112B                     0.0s
=> [internal] load metadata for docker.io/library/python:3.9-slim 5.3s
=> [auth] library/python:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore                        0.7s
=> => transferring context: 2B                          0.0s
=> [1/3] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13c 19.3s
=> => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf 0.8s
=> => sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 29.78MB / 29.78MB 13.2s
=> => sha256:ea56f685404adf81680322f152d2cfec62115b30dda481c2c450078315beb508 251B / 251B 0.6s
=> => sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881d 1.29MB / 1.29MB 2.7s
=> => sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 13.88MB / 13.88MB 10.3s
=> => extracting sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 0.9s
=> => extracting sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881d 0.5s
=> => extracting sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 0.7s
=> => extracting sha256:ea56f685404adf81680322f152d2cfec62115b30dda481c2c450078315beb508 0.4s
=> [internal] load build context                        1.7s
=> => transferring context: 55B                         0.0s
=> [2/3] WORKDIR /app                                6.0s
=> [3/3] COPY app.py .                                1.1s
=> exporting to image                                  6.4s
=> => exporting layers                                    3.3s
```

```
C:\Users\basil\Downloads\python_app>docker run python_app
Hello, World!
```

The screenshot shows the Visual Studio Code interface with the Dockerfile editor open. The Explorer pane on the left shows the project structure with 'PYTHON_APP' containing 'app.py' and 'Dockerfile'. The Dockerfile editor shows the following content:

```
Dockerfile > ...
1 FROM python:3.9-slim AS builder
2 WORKDIR /app
3 COPY app.py .
4
5 FROM python:3.9-alpine
6 WORKDIR /app
7 COPY --from=builder /app /app
8 CMD ["python", "app.py"]
```

```
C:\Users\basil\Downloads\python_app>docker login
Authenticating with existing credentials... [Username: ajxshadow]
```

Info → To login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded

```
C:\Users\basil\Downloads\python_app>docker tag python_app:v2 ajxshadow/python_app:v2
```

```
C:\Users\basil\Downloads\python_app>docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
ajxshadow/python_app:v2	bb6b69a41bb	79.7MB	1.7MB	
docker.n8n.io/n8nio/n8n:latest	0a65e6e5995c	1.68GB	278MB	
image_name:latest	0a43d41dd7fb	237MB	62.9MB	
mysql:latest	24e450bbd24f	1.27GB	283MB	U
nginx:latest	7150b3a39203	240MB	65.8MB	U
postgres:15-alpine	fceb6f86328c	392MB	109MB	
postgres:15.17-trixie	f30e3de0ac9c	627MB	158MB	
python_app:latest	4c0d24aa1341	183MB	45MB	U
python_app:v2	bb6b69a41bb	79.7MB	1.7MB	
ubuntu:latest	186072bba1b2	119MB	31.7MB	

```
C:\Users\basil\Downloads\python_app>docker push bb60b69a41bb
Using default tag: latest
The push refers to repository [docker.io/library/bb60b69a41bb]
tag does not exist: bb60b69a41bb:latest
```

```
C:\Users\basil\Downloads\python_app>docker push ajxshadow/python_app:v2
The push refers to repository [docker.io/ajxshadow/python_app]
61bcefa7ad01: Pushed
9806928fa0da: Pushed
2d35ebb057d9: Pushed
27711589a568: Pushed
946d712b015f: Pushed
f07ebfac7cc0: Pushed
c9008197b8dd: Pushed
v2: digest: sha256:b6b069a41bbe3deba49610233cc6122fb9462751177121915619fd3dec07a1 size: 856
```